



MoldGuard: IoT-Based Mold Risk Prevention & Auto-Dehumidification System

Software Development Proposal

Abstract

An IoT-integrated, browser-based mold risk prevention and detection system where authorized users monitor real-time humidity, temperature, and air quality (VOC) conditions, receive predictive mold risk alerts calculated through a proprietary Mold Risk Index, and benefit from closed-loop automatic dehumidification that activates before mold-conducive conditions are reached.

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Executive Summary

Mold growth in residential spaces is a well-documented consequence of prolonged high humidity and poor ventilation, particularly during the rainy season, and is closely linked to structural damage as well as respiratory and allergy-related health risks. However, most households have no way of knowing that conditions are becoming favorable for mold until visible growth or odor already appears. Even where dehumidifiers are used, most operate reactively — either as passive moisture absorbers with no monitoring capability, or as basic smart devices that respond only to a single humidity threshold without accounting for exposure duration or temperature.

To solve this problem, our team proposes **MoldGuard V1.0** — a compact, IoT-based mold risk prevention and detection system. MoldGuard combines temperature, humidity, and volatile organic compound (VOC) sensing with a proprietary Mold Risk Index algorithm to predict rising mold risk before conditions become critical, automatically activating a Peltier-based dehumidification unit in a closed-loop response. A companion web dashboard and mobile app give users real-time status, historical trend analytics, and manual override control. This end-to-end architecture shifts mold management from reactive cleanup to proactive, automated prevention.

1. Project Overview

General Objective:

To develop MoldGuard V1.0, a compact IoT-based mold risk prevention and detection device with an accompanying web dashboard and mobile app that automatically monitors and mitigates mold-conducive conditions.

Specific Objectives:

1. To design a Mold Risk Index algorithm that calculates real-time mold growth risk using relative humidity, temperature, and sustained exposure duration, rather than a simple humidity threshold.
2. To integrate a volatile organic compound (VOC) sensor into the sensor array to detect microbial off-gassing indicative of existing mold growth, ahead of visible signs.
3. To implement closed-loop automatic dehumidification using a compact TEC1-12706 Peltier thermoelectric module, triggered directly by the Mold Risk Index without requiring manual user action.
4. To develop a web dashboard providing historical humidity, temperature, and VOC trend analytics, with support for multi-room or multi-unit monitoring.
5. To develop a companion mobile app for real-time push notifications, live device status, and manual override of the dehumidification unit.

System Features:

1. **Mold Risk Index Engine:** Real-time risk scoring based on humidity, temperature, and exposure duration — going beyond simple threshold triggers used by commercial dehumidifiers.
2. **Early Mold Detection:** VOC sensing to flag existing microbial growth before it becomes visible or noticeable by smell.
3. **Closed-Loop Auto-Dehumidification:** Automatic activation of the Peltier-based dehumidifying unit once risk thresholds are met — no manual intervention required.
4. **Web Dashboard & Analytics:** Historical trend logging and a multi-room or multi-unit view for larger households, dormitories, or property managers.
5. **Mobile App Alerts & Control:** Real-time push notifications and manual override of the dehumidification unit from anywhere.

2. Statement of the Problem

The project addresses the lack of proactive, automated mold prevention tools available to renters and households, particularly during the rainy season. Specific obstacles solved include:

- Inability to detect rising mold risk before visible mold growth or odor already appears.
- Existing dehumidifiers relying on simple humidity threshold triggers that ignore exposure duration and temperature, resulting in delayed or ineffective response.
- Passive moisture absorbers that offer no monitoring, data, or automation whatsoever.
- Lack of affordable, vendor-agnostic monitoring options for renters and small property owners compared to enterprise-grade building monitoring systems.

3. Technical Obstacles

- **Sensor Fusion & Risk Modeling:** Combining humidity, temperature, and VOC readings into a single reliable Mold Risk Index without excessive false positives or negatives.
- **Thermal Management:** Efficiently dissipating heat from the Peltier module's hot side within a compact enclosure without compromising unit size.
- **Condensate Handling:** Safely collecting and containing water extracted from the air without leaks or overflow in a compact housing.
- **Real-Time Synchronization:** Maintaining reliable low-latency data sync between the ESP32 microcontroller, backend server, web dashboard, and mobile app.

4. Industry and Market Risks

Existing commercial dehumidifiers and smart humidity devices are designed primarily around general comfort or simple humidity-percentage control, not mold-specific risk modeling, and typically operate within closed, proprietary mobile-app ecosystems without a web-based analytics layer. Passive absorber products, meanwhile, provide no monitoring or automation whatsoever.

System	Functions	Pricing Model	Drawbacks
Passive Moisture Absorbers	Chemical moisture absorption, no sensors	Low one-time cost	No monitoring, no data, no automation
Commercial Smart Dehumidifiers	App-controlled humidity threshold triggers	Mid-to-high retail hardware cost	No mold-specific modeling; locked to vendor app; no web dashboard
MoldGuard V1.0	Mold Risk Index (humidity + temperature + duration + VOC), closed-loop auto-dehumidification, web dashboard, mobile app	Fixed development fee	Requires dedicated power source and periodic condensate tray emptying

5. Budgetary Risks

Commercial smart dehumidifiers often come with a high retail hardware cost and lock users into a single vendor's app ecosystem. MoldGuard utilizes open hardware components (ESP32, TEC1-12706 Peltier module, off-the-shelf sensors) and free/open-source software services, providing a single fixed-cost development structure with full client ownership and no recurring subscription fees.

Technology Requirement (Software and Hardware)

6. Hardware

Component Category	Development /Server Station	MoldGuard Device (Node)	Specifications
Microcontroller / CPU	Intel Core i5 / AMD Ryzen 5 (8GB+ RAM)	ESP32-WROOM-32 Dev Board	Dual-core 240MHz, WiFi + Bluetooth, 30+ GPIO pins, 3.3V logic
Power System	Standard Utility Power (220V AC)	12V/2A DC Power Adapter + Buck Converter	Steps down to 5V (ESP32) and 12V (Peltier module); barrel jack input
Thermal Components	N/A	TEC1-12706 Peltier Module	12V, 6A max, up to 92W heat transfer; paired with aluminum heatsink (40x40x11mm) and 12V DC brushless fan (40mm, ~0.15A) for hot-side heat dissipation
Sensing Temp/Humidity –	Development PC Interface	SHT31-D Sensor (or DHT22 as budget alternative)	I2C interface, ±2% RH accuracy, ±0.3°C temp accuracy, range: 0-100% RH, -40°C to 125°C

Sensing – VOC	N/A	BME680 (recommended) or MQ-135 (budget option)	BME680: I2C, integrated temp/humidity/pressure/gas sensor, IAQ index output. MQ-135: analog output, 10-1000ppm NH3/VOC detection range, requires 24-48hr burn-in for calibration
Actuators & Relays	N/A	1-Channel 5V Relay Module or IRF520 MOSFET Driver	Relay: 10A/250VAC rated, opto-isolated. MOSFET: for low-voltage DC switching of Peltier circuit, logic-level gate
Enclosure	N/A	Custom compact housing (ABS plastic or 3D-printed PLA)	Approx. 15cm x 10cm x 12cm; ventilated intake/exhaust slots; removable condensate tray (~50-100ml capacity); wall-mount bracket or tabletop base
Connectivity	Router/WiFi Access Point	Built-in ESP32 WiFi module	802.11 b/g/n, 2.4GHz only; connects to backend via WiFi for real-time sync
Data Storage (onboard)	N/A	ESP32 onboard flash (4MB)	Local buffering of sensor readings during connectivity loss, syncs once reconnected

7. Software

Development Environment

- Operating System: Windows 10/11, macOS, or Linux
- IDE: Visual Studio Code, Arduino IDE / PlatformIO
- Firmware Language: C++ (for ESP32 logic, sensor polling, and Peltier/relay switching)
- Version Control: Git & GitHub

Application Layer & Client Interfaces

- **UserUI (Web Dashboard):** Built using Next.js, React, TypeScript, and Tailwind CSS. This dashboard hosts the Mold Risk Index Engine view, displaying real-time risk scores and historical humidity/temperature/VOC trend charts, with manual override controls for the dehumidification unit.
- **Mobile App:** Built using Flutter or React Native, delivering real-time push notifications for rising mold risk and live device status alongside manual override controls.
- **Real-Time Event Bus:** Utilizes WebSockets via Socket.io to stream low-latency bidirectional sensor readings, risk score updates, and dehumidification trigger events between the physical device and the dashboard/app.

Design and Testing Tools

- Figma
- Postman
- Google Chrome / Brave / Microsoft Edge / Mozilla Firefox

Cloud Services & Backend

- **Runtime Environment:** Node.js backend handling the Mold Risk Index computation and real-time state synchronization.
- **API Framework:** Express.js handling REST API pipelines for sensor telemetry, historical logging, and device control commands.
- **Database:** PostgreSQL serves as the primary data store for humidity/temperature/VOC historical logs, risk score history, and device event records.

Hosting and Deployment

- **Frontend:** Vercel hosts the Next.js Admin UI for scalable browser access.
- **Backend:** Railway or Render securely hosts the Node.js/Express application.
- **Database:** Neon hosts the PostgreSQL database in a secure cloud-based, serverless environment.

8. Milestones and Reporting

1. Communication
2. Planning
3. Modeling
4. Construction
5. Deployment

Milestone	Tasks	Reporting	Days	Date
Communication	Gather information on mold growth science and IoT sensor requirements; Define MoldGuard system requirements; Build initial sensor and Peltier prototypes; Prioritize features (Mold Risk Index vs. VOC detection); Generate and evaluate architecture alternatives; Review recommendations with adviser/panel	To-be-agreed upon	112 hours	To-be-agreed upon
Planning	Design the Mold Risk Index algorithm and calibration logic; Design the web dashboard and mobile app architecture; Design ESP32 firmware and sensor interfacing; Design and integrate the database schema; Prototype Peltier dehumidification control and condensate handling		280 hours	

Milestone	Tasks	Reporting	Days	Date
Modeling	Verify and test Mold Risk Index accuracy against controlled humidity/temperature/duration scenarios; Test VOC sensor sensitivity and response time; Load sample historical data for dashboard analytics testing	To-be-agreed upon	112 hours	To-be-agreed upon
Construction	Build backend API and real-time sync pipelines; Assemble and integrate the compact enclosure; Install firmware and deploy software stack; Conduct calibration and bug fixes; Enhance Mold Risk Index thresholds based on test results		168 hours	
Deployment	Final user training and documentation handover; On-site or remote installation support		50 hours	
Total Duration			722 hours	

9. Deployment

The software will be distributed in the following manner/s:

- **MoldGuard Device:** Compact tabletop or wall-mounted physical installation near humidity-prone areas (e.g., bathroom, closet, basement).
- **Web Dashboard:** Cloud-hosted browser access optimized for desktop and mobile browsers.

10. Testing

- **Mold Risk Index Accuracy Testing:** Validating risk scoring against controlled humidity, temperature, and duration test scenarios.
- **VOC Detection Testing:** Verifying sensor sensitivity and response time to simulated microbial off-gassing.
- **Dehumidification Response Testing:** Confirming automatic Peltier activation occurs correctly once risk thresholds are met.
- **Real-Time Sync Testing:** Stress-testing data sync between the device, backend, web dashboard, and mobile app.

11. Documentation

The document provided will be as follows:

- Software Development Proposal
- Hardware Circuit Schematics & Wiring Manual
- System Operations & API Technical Documentation

12. Contact Us

Our Team

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If you would like to proceed with our proposal then you can sign the first page and return a copy to us by fax, email or post.

In any case please feel free to call us to discuss the quote, request more information or for any other reason.

We look forward to hearing from you soon!